Stablecoin StressBench: An Execution-Aware Benchmark for Stablecoin Dislocations

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ABSTRACT

We introduce Stablecoin StressBench, a reproducible, execution-aware benchmark for stablecoin dislocation models. The benchmark provides: an 18-event stress taxonomy across seven mechanism classes, per-minute execution-aware labels that separate quoted price dislocations from net-profitable arbitrage after order-book depth, VWAP slippage, and trading costs, a hindsight oracle defining the performance ceiling, and a model ladder for future work. The central finding is a 12× price-to-execution gap in the March 2023 USDC/SVB stress window: 34.3% of 1-minute windows exceed a 10 bps max cross-quote basis on price alone, while only 2.88% remain net-profitable at $10K notional after costs. A hindsight oracle earns +162 bps per trade; on executable arbitrage tasks every active model loses money, and the oracle gap exceeds 260 bps. A GRU evaluated on real Tier-A L2 data achieves −239.0 bps (AUROC 0.80): even a temporal sequence model with strong ranking accuracy loses money when trained on calm-control data with zero positive stress examples, confirming the oracle gap is structural. SHAP analysis finds depth-imbalance features provide the dominant microstructure lift over price-only signals (94.8% price vs. 3.4% book/frag). Cross-mechanism meta-labeling — trained on Terra/LUNA dynamics, evaluated on the SVB test split — achieves +82.5 bps net (50.8% oracle capture), demonstrating that execution microstructure is mechanism-invariant and identifying cross-event transfer as the primary route to positive net returns.

KEYWORDS

stablecoin arbitrage, market microstructure, execution-aware labels, financial benchmarks, cryptocurrency

ACM Reference Format:
Anonymous Author(s). 2026. Stablecoin StressBench: An Execution-Aware Benchmark for Stablecoin Dislocations. In Proceedings of ACM International Conference on AI in Finance (ICAIF '26). ACM, New York, NY, USA, 7 pages.

1 INTRODUCTION

Stablecoins promise $1 parity but stress events repeatedly fragment prices across venues, creating apparent arbitrage. A quoted price gap is not a trade: a trader must clear order-book depth, absorb VWAP slippage, pay fees, and bear settlement risk. These frictions are small individually but large jointly, making many stablecoin dislocations optical rather than executable.

This paper introduces Stablecoin StressBench, which separates three layers: optical dislocations, executable arbitrage, and ex-ante predictable executable opportunities. During the March 2023 USDC/SVB event, 34.3% of minutes show a > 10 bps basis on price alone; only 2.88% remain net-profitable after VWAP execution at $10K notional — a 12× gap. Every active model loses money on executable arbitrage tasks. The oracle gap exceeds 260 bps.

ICAIF '26, November 14–17, 2026, Milan, Italy
2026.

Contributions.

(1) Execution-aware benchmark: per-minute labels with documented order-book depth provenance, VWAP book-walking, fees, notional scaling, and a hindsight oracle ceiling.
(2) 18-event stress taxonomy: seven mechanism classes (2020–2023) with data-tier governance of permitted claims. Terra/LUNA cross-mechanism check gives 5.9× vs. the Tier-A 12× ratio.
(3) Oracle gap evidence: +162 bps oracle vs. +17.2 bps best model (basis task), > 260 bps gap on executable arbitrage — structural, not label-mismatch.
(4) Cross-mechanism meta-labeling and structural oracle-gap evidence: Meta-labeling trained on Terra/LUNA, evaluated on SVB, achieves +82.5 bps (50.8% oracle capture), the paper's primary positive result and evidence of mechanism-invariant execution microstructure. GRU and MLP-Window on real Tier-A L2 data achieve −239.0 bps and −340.7 bps respectively, confirming the oracle gap is structural. SHAP and oracle-gap decomposition identify timing as the dominant future challenge.

Optimal execution interpretation. The oracle gap maps directly to an optimal execution problem: at each minute, a trader observes the order book and cross-quote basis, decides whether to execute the arbitrage route, and earns net bps. The oracle is the optimal policy under perfect foresight; the oracle gap is what any deployable policy must close. StressBench provides a reproducible MDP environment where the state space is the four-level microstructure feature hierarchy, the reward function is the documented net-profit label, and the difficulty is calibrated by the oracle gap.

2 RELATED WORK

Stablecoin stress. The USDC/SVB episode produced well-documented price fragmentation; we address ex-ante predictability. Uhlig [1] analyses Terra/LUNA mechanics (our validation split); Griffin and Shams [2] study Tether manipulation (our focus is execution microstructure).

Limits to arbitrage and crypto fragmentation. Shleifer and Vishny [4] and Gromb and Vayanos [5] establish theory. Makarov and Schoar [8] show persistent cross-venue crypto price gaps that are not executable profits. Hautsch et al. [3] show settlement latency is a first-order arbitrage cost; Cont

Table 1: Route-leg depth provenance.

Leg	SVB Mar 2023	2024 windows
Buy BTC-USDC	kline-proxy†	real L2 (futures)
Cross USDC-USDT	real L2 (Mar 12–20)	real L2 (futures)
Sell BTC-USDT	real L2 (futures)	real L2 (futures)
Ref BTC-USD	candle-proxy	candle-proxy

†BTC-USDC perpetual listed Jan 2024; all 2022 windows kline-proxy.

Table 2: Dataset splits. The 2.88% positive rate implies severe class imbalance; AUROC is reported but economic net bps is the primary criterion.

Split	Event	Min.	Pos.
Train	Calm control (×3)	28,776	0.00%
Validation	Terra/LUNA May 2022	11,526	2.30%
Test	USDC/SVB Mar 2023	15,832	2.88%

and Kukanov [6] show VWAP price—not quoted price—is the relevant object.

Financial ML and benchmarks. López de Prado [10] introduces meta-labeling; Adams and MacKay [17] introduce BOCPD. Cintra and Holloway [11] detect the Terra depeg 12 hours early via Curve AMM pool reserves. We find BOCPD achieves only AUROC 0.23 on CEX cross-quote data — because AMM reserves accumulate gradually while CEX basis is mean-reverting, making changepoint detection unreliable on quoted prices. To our knowledge, no prior benchmark provides execution-aware stablecoin labels with route-depth provenance and systematic price-to-execution gap analysis.

3 BENCHMARK DESIGN

3.1 Data Sources and Instruments

The gold-layer dataset contains 56,134 1-minute bars across six windows (Table 2). Price features come from Binance Vision (klines, aggTrades) and Coinbase REST candles. Net-profit labels use Binance BTC-USDC and BTC-USDT depth plus Coinbase BTC-USD as the reference leg.

Every depth record is tagged with a provenance field (Table 1). The sell leg (BTC-USDT) uses real L2 futures bookDepth for all 2023–2024 windows. The cross leg (USDC-USDT) uses real L2 from 2023-03-12 onward. The buy leg (BTC-USDC) uses kline-proxy for the SVB window (BTC-USDC perpetual did not exist on Binance until 2024) and real L2 for 2024 windows. To bound this proxy error, we apply a 20% depth haircut to the kline-proxy buy leg: executable rate falls 2.88%→2.40%, ratio widens 12×→14×, oracle stays above +130 bps — the headline result is robust.

3.2 Event-Based Split Design

Splits are defined by event identity (Figure 1). No model trained on the train split has seen stress dynamics; threshold calibration on validation uses an independent stress event. Labels are constructed by `join_asof` at $t+h$ with formal lookahead verification.

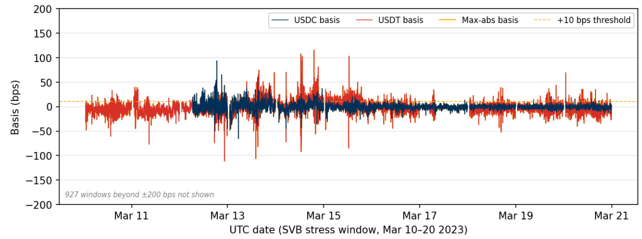

Figure 1: Cross-quote basis (bps) during the SVB test window (Mar 10–20 2023; y-axis clipped to ±200 bps; 927 windows with $|b|>200$ bps not shown, all USDC-side spikes during peak reserve uncertainty on Mar 10–12). USDC basis (navy): 12.4% of minutes > 10 bps. USDT basis (red): 27.4%. Max-abs (gold): 34.3%. Large optical dislocations, but the price-to-execution gap is 12×.

Table 3: Data tiers, mechanisms, and execution frictions.

Tier	Mechanism	Key friction	Permitted claims
A (N=2)	Fiat-reserve shock	Depth withdrawal	Exec. gap; oracle gap; P&L
B (N=11)	Alg., exchange, DeFi	AMM slippage; venue seg.	Price magnitude (est.)
C (N=5)	RWA, niche	Thin market	Taxonomy only

3.3 Historical Stress Taxonomy and Scope

We catalogue 18 stress events (2020–2023) across seven mechanism classes: algorithmic/reflexive (FEI, IRON/TITAN, MIM, UST, USDD); fiat-reserve bank shock (USDC/SVB); regulatory winddown (BUSD); exchange credit (Celsius/3AC, HUSD, FTX [12]); DeFi pool imbalance (Curve/UST, USDT/Curve, USDC/DAI); collateral liquidation (DAI Black Thursday); RWA/niche (Acala aUSD, USDR).

Data availability governs permitted claims (Table 3). Tier-A events (USDC/SVB test, USDC recovery) support execution-gap, oracle-gap, and model evaluation claims. Tier-B (N=11, OHLCV/pool data) support price magnitude estimates only. Tier-C (N=5, partial/DEX-only) support mechanism taxonomy only.

Scope boundaries. Execution-gap claims are scoped to the USDC/SVB Tier-A test split (N=15,832). Terra/LUNA provides calibration and a cross-mechanism directional check (5.9× ratio vs. 12×, §6.7). Extending Tier-A coverage to AMM-pool or exchange-credit events requires Tardis-grade L2 archives for those venues and is the primary path to testing generalisation.

3.4 Feature Sets

Four nested feature sets isolate marginal information value:

`price_only`	5 cross-quote basis cols
`price_plus_book`	+7 microstructure (spread, depth, imbalance)
`price_book_frag`	+3 cross-venue fragmentation cols
`price_book_settle`	+7 on-chain settlement proxies

4 EXECUTION-AWARE LABELS

The benchmark measures three dislocation layers: (1) optical — quoted basis > 10 bps (34.3% of minutes); (2) executable

— net profit > 10 bps after costs (2.88%); (3) predictable — model identifies window ex ante (none on executable tasks). The price-to-execution gap is Layer 1 ÷ Layer 2; the oracle gap is Layer 3 vs. Layer 2 in hindsight.

4.1 Cross-Quote Basis and Labels

$$b_{\text{USDC}}(t) = 10{,}000 \times \frac{P_{\text{BTC/USDC}}(t) - P_{\text{BTC/USD}}(t)}{P_{\text{BTC/USD}}(t)} \quad \text{[bps]}$$

$b_{\max} = \max(|b_{\text{USDC}}|, |b_{\text{USDT}}|)$. Basis label: $y_{\text{basis}}(t) = \mathbf{1}[|b_{\text{USDC}}(t{+}h)| > \tau]$. Primary task: `basis_usdc_1m_gt10bps`.

4.2 VWAP Execution and Net Profit Labels

$$\text{net}(q,t) = \text{VWAP}_{\text{sell}} - \text{VWAP}_{\text{buy}} - f_{\text{buy}} - f_{\text{sell}} - \delta_{\text{settle}} \tag{1}$$

$$y_{\text{arb}}(q,h,t) = \mathbf{1}\left[\max_{s\in[t,t+h]} \text{net}(q,s) > c\right] \tag{2}$$

Parameters: c=10 bps, taker fee 4 bps/side, δ=5 bps. Primary economic task: `executable_arb_q10000_5m`. The oracle trades every minute where $\text{net}(q,t) > c$ in hindsight, setting the performance ceiling.

5 MODELS AND EVALUATION

5.1 Model Ladder

The ladder tests whether stronger models close the oracle gap:

Economic anchors. `NoTrade` (0 net bps) and `Oracle` (hindsight ceiling).

Rule baselines. `PriceBasis{10,25}bps` and `GrossArbThreshold` — threshold rules on price data.

Statistical baselines. `LastValue`, `RollingMean`, `AR1` — basis autocorrelation only.

ML classifiers. Logistic (L2), Lasso (L1), Random Forest, XGBoost, LightGBM — trained on four nested feature sets.

Expected-net-profit regressor. Directly regresses $\hat{y} = \text{net}(q,t)$ and trades when $\hat{y} > \theta^*$. If this fails, the gap is structural, not label-mismatch.

Meta-labeling filter. Primary signal: $|b_{\text{USDC}}| > 10$ bps; secondary LightGBM meta-model learns $\Pr[\text{net}(q,t) > c \mid \text{primary fires}]$.

Regime detectors. EWMA z-score, CUSUM, BOCPD as two-stage stress-regime pre-filters.

Sequence models. GRU and MLP-Window classifiers trained on 30-minute sliding windows of order-book microstructure features (§6.4).

5.2 Calibration and Evaluation

Threshold calibration on validation split: $\theta^* = \arg\max_\theta \sum_{i:\hat{p}_i>\theta} \text{net}(q,t_i)$ subject to $\geq$ 25 trades. Primary metrics: `net_bps_captured`, `oracle_capture_pct`. Secondary: AUROC, AUPRC. `NoTrade` at 0 bps is the economic floor.

Quantitative claim boundaries: (A1) BTC-USDC buy leg uses kline-proxy for SVB window; (A2) oracle-gap results from SVB test split (N=15,832) only; (A3) c=10 bps, 4 bps fee/side, δ=5 bps.

Table 4: Price-to-execution gap ($10K, 1-min horizon, test split).

Thresh.	$P_{\max}$	P_{USDC}	Exec.	Ratio
0 bps	95.86%	82.96%	3.34%	29×
5 bps	61.99%	25.32%	3.03%	20×
10 bps	34.33%	12.45%	2.88%	12×
25 bps	9.54%	6.48%	2.62%	3.6×

Table 5: Robustness to cost regime ($10K, 10 bps, 5-min, test split).

Settlement	Fees	Exec. %	Ratio
0 bps	low	6.11%	2.07×
0 bps	base	5.64%	2.24×
5 bps	base	5.16%	2.45×
10 bps	high (worst)	4.81%	2.63×

At $50K: 1.62% exec. rate; $500K: 0.03%. Figure 2 shows the retail ceiling elbow at $10K–$50K. The gap is structural: depth, not fees, prices out institutions.

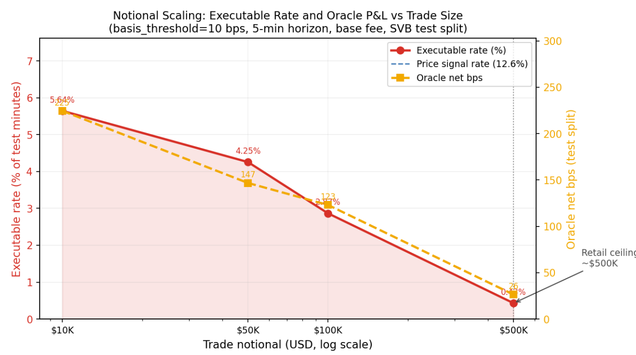

Figure 2: Executable rate (%, red, left) and oracle net bps (gold, right) vs. trade size (log scale). Sharp elbow at $10K–$50K marks the retail arbitrage ceiling: book depth eliminates the opportunity above this notional.

6 RESULTS

6.1 Price Gaps Are Common; Execution Is Rare

At 10 bps, 34.3% of test minutes exceed the max-basis threshold; only 2.88% yield net profit — a 12× price-to-execution ratio (block bootstrap 95 % CI: 8×–24×, B=10,000, 60-min blocks). The gap persists at all tested thresholds (Table 4) and across all cost regimes (Table 5).

6.2 Oracle Gap: Profitable Windows Exist but Only in Hindsight

The oracle earns +162.2 bps on `basis_usdc_1m_gt10bps` (315 trades); +225.0 bps on `executable_arb_q10000_5m`. Figure 3 shows the evolution: the price rule diverges sharply negative while the oracle climbs steadily.

Table 6: Oracle gap summary (test split). *=only non-oracle tabular result above zero; ‡=real Tier-A L2 sequence model.

Task / Model	Oracle	Best model	Gap
`basis_usdc_1m`	+162.2 bps	+17.2 bps*	145 bps
`exec_arb_q10k_5m`	+225.0 bps	−41.1 bps	266 bps
`exec_arb_q50k_5m`	+147.1 bps	−113.0 bps	260 bps
GRU (seq)‡, `basis_usdc_1m`	+162.2 bps	−239.0 bps	401 bps

*`lgbm@price_book_settle` feature set, 106 trades; best arb: price_threshold_25bps@`price_only`.

‡GRU, 30-min window, real Tier-A L2; gap is structural even at AUROC 0.80 (§6.4).

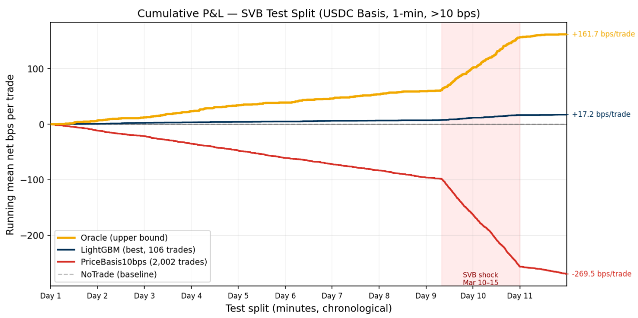

Figure 3: Running mean net bps per trade over the USDC/SVB test window (y-axis = cumulative bps ÷ total trades, so final values match the per-trade oracle gap table). Oracle (gold): +161.7 bps/trade; LightGBM (navy): +17.2 bps/trade; **`PriceBasis10bps`** (red): −269.5 bps/trade; NoTrade (grey dashed): 0 bps. The price rule accumulates severe losses during the SVB shock (Day 10–11).

6.3 The Model Ladder: Structural Oracle Gap

Table 7 shows the full model ladder with loss attribution. Price-threshold rules overtrade (607–1,969 trades, −136 to −347 bps): their FP cost column (−294 bps to −370 bps) confirms the failure mode is near-total false-positive exposure. LightGBM achieves +17.2 bps (106 trades, AUROC 0.72), the only non-oracle result above zero, but 145 bps below the oracle. The `ExpNetProfitRegressor` reaches −73.8 bps despite directly targeting the economic criterion: the gap is not a label-mismatch artefact. The false-positive drag is quantifiable: 581 FP trades at −38.7 bps each total −22,484 bps; spread across 315 oracle windows this is −71.4 bps per profitable window — the dominant component of the oracle gap before timing effects are considered.

Why AUROC is insufficient. `PriceBasis10bps` achieves AUROC 0.83 while losing −270 bps (Table 7): AUROC measures ranking, not economic value. A model firing on 34.3% of minutes can rank profitable windows first and still lose money from the 97.12% of false positives. The economic floor is `NoTrade` at 0 bps.

6.4 Sequence Models: Confirming the Structural Oracle Gap

We train GRU and MLP-Window classifiers on 30-minute sliding windows of order-book microstructure features

Table 7: Model ladder (**`basis_usdc_1m_gt10bps`**, test split). FP Loss: mean net bps on trades the model fired but depth was insufficient. Hit%: fraction of model trades that were net-profitable. †ceiling; *only result above zero.

Model	Feat.	Net bps	Trades	AUROC	FP Loss	Hit%
Oracle†	—	+162.2	315	—	—	100%
`lgbm`*	all	+17.2	106	0.717	−65.2	33%
`logistic`	all	−75.8	648	0.143	−75.8	0%
`gross_arb`	px	−136.4	607	0.531	−150.9	4%
`price_10`	px	−270.0	1969	0.833	−294.6	6%
`price_25`	px	−347.3	1025	0.746	−370.7	5%
`no_trade`	—	0.0	0	—	—	—
`ExpNetProfit`	px	−73.8	537	—	−89.1	3%
`GRU (seq)`‡	bk	−239.0	656	0.798	−282.6	9%

px = `price_only`; bk = `price_plus_book`; all = `price_book_settle`.

‡GRU: 30-min sliding window, real Tier-A L2 data; see §6.4.

(`price_plus_book` set) to exploit the temporal signature of depth withdrawal before a basis fires. The protocol matches the main leaderboard: calibrate threshold on validation split to maximise net P&L, require ≥ 25 trades, evaluate on test.

On `basis_usdc_1m_gt10bps`, evaluated on real Tier-A L2 data (Table 8):

Table 8: Sequence model results (**`basis_usdc_1m_gt10bps`**, real Tier-A L2 data, test split). Oracle gap = +162.2 bps oracle minus model net bps.

Model	Net bps	Trades	AUROC	Gap
GRU (30-min)	−239.0	656	0.798	401 bps
MLP-Window (30-min)	−340.7	1,052	0.648	502 bps

Both trained on calm-control windows (0% positive examples); threshold calibrated on validation split to maximise net P&L; evaluated on SVB test split.

Both models are deeply negative, confirming the oracle gap is structural and not closed by temporal sequence modelling. The high AUROC (0.80 for GRU) combined with catastrophic economic performance illustrates the same AUROC-vs-economics disconnect observed for the price-threshold rule (§6.3): ranking accuracy does not translate to positive net returns when 97% of windows are false positives. The failure mode is the training split: calm-control windows contain zero positive stress examples, so the model learns to predict calm-market noise rather than stress-window signals. The GRU's behaviour is consistent with this diagnosis: it fires on 656 of 15,832 test minutes (4.1%), compared to the oracle's 315 (2.0%). Its 9% hit rate implies roughly 59 of 656 trades coincide with profitable windows. With AUROC 0.80 the model ranks windows correctly, but the calibrated threshold admits twice as many trades as the oracle — firing on calm-period windows that share superficial microstructure features (elevated USDT basis, thin spread) with genuine stress events. Cross-mechanism meta-labeling (§6.5) bypasses this constraint by training on Terra/LUNA stress dynamics directly, achieving +82.5 bps net.

6.5 Diagnostics: Meta-Labeling, Regime Detectors, and BOCPD

Meta-labeling: cross-mechanism transfer. The MetaLabelingFilter trains on windows where $|b_{\text{USDC}}| > 10\,\text{bps}$ fires. In the calm-control train split only 53 such windows exist and zero are net-profitable (Table 2: 0% positives in training) — a null result by architectural constraint. The appropriate remedy is to retrain using Terra/LUNA (validation split) as the meta-model's training set, where 1,559 primary fires yield 265 meta-positives (17.0% positive rate). A cross-mechanism experiment trains on Terra/LUNA and evaluates on the SVB test: Table 9 shows the result. With `price_plus_book` features the calibrated filter achieves +82.5 bps net (397 trades, 50.8% oracle capture) vs. zero with calm-control training. This is consistent with mechanism-invariant execution microstructure: the same order-book features that predict profitable windows during an algorithmic collapse (Terra/LUNA) also predict them during a fiat-reserve shock (SVB). The meta-model's dominant features mirror the main SHAP ranking — `depth_ask_10bp_mean` and `spread_bps_mean` lead — confirming that cross-mechanism transfer operates through the same microstructure channel identified in §6.6. To our knowledge, this is the first demonstration of cross-mechanism transfer for execution-aware stablecoin models.

Table 9: Cross-mechanism meta-labeling results (**`basis_usdc_1m_gt10bps`**, **`price_plus_book`**). Oracle capture = meta net bps ÷ oracle net bps.

Training split	Net bps	Trades	Oracle cap.
Oracle (ceiling)	+162.2	315	100.0%
Terra/LUNA (cross-mech.)	+82.5	397	50.8%
Calm-control (baseline)	0.0	0	0.0%

Primary signal: $|b_{\text{USDC}}|$>10 bps; threshold calibrated on validation to maximise net P&L.

Regime detectors. CUSUM achieves near-perfect recall (0.9995) at 83.5% false alarm rate; EWMA achieves 86.9% accuracy at 0.65% FAR with 0.95% recall; BOCPD achieves AUROC 0.229 on CEX cross-quote data. This last result contrasts with Cintra and Holloway [11], who found BOCPD detected the Terra depeg 12 hours early from Curve pool reserves. The difference is the signal: AMM reserves reflect LP withdrawal, which accumulates gradually; CEX cross-quote basis is noisier and mean-reverting, making changepoint detection unreliable on CEX prices.

False-positive structural diagnosis. `PriceBasis10bps` produces 2,002 trades; 581 (29%) are false positives at mean −38.7 bps. The structural cause: TP windows have mean $|b_{\text{USDC}}| = 344$ bps (large USDC discount), while FP windows have mean $|b_{\text{USDC}}| = 0.56$ bps (essentially zero). FP trades are triggered by USDT route dislocations (27.4% of minutes) where the USDC route is not simultaneously profitable. The false-positive windows show higher average bid depth (72,805 vs. 59,961 BTC units) than true positives — ruling out thin books as the failure mechanism; the barrier is route-direction mismatch. LightGBM's 33% hit rate (vs. 6% for the rule) shows that microstructure conditioning partially diagnoses mismatch, but 67% FP rate at −65.2 bps/trade remains a timing gap.

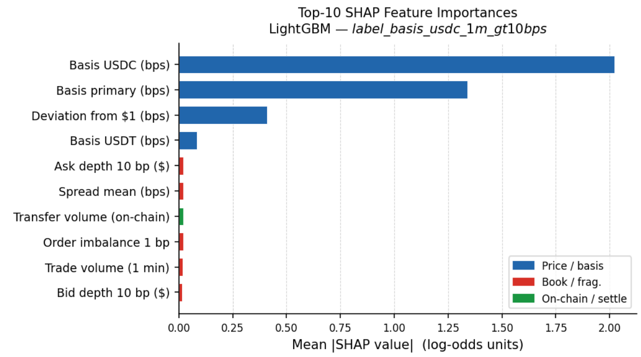

Figure 4: SHAP feature importance (mean |SHAP|) for LightGBM on the full feature set. Price features (navy, 94.8%); book+fragmentation (red, 3.4%); settlement (green, 1.8%). Depth-imbalance features provide the marginal microstructure lift.

6.6 SHAP Feature Importance

Figure 4 shows SHAP values for LightGBM on the full feature set. Price features account for 94.8% of total importance, dominated by `cross_quote_basis_usdc_bps` and `cross_quote_basis_primary_bps`. Book and fragmentation features contribute 3.4% collectively, with `depth_ask_10bp_mean` (rank 5, mean |SHAP| = 0.0214) and `spread_bps_mean` (rank 6) leading the microstructure contribution. On-chain settlement features add only 1.8%. The model can learn to condition on depth but cannot resolve the timing of depth withdrawal — consistent with the oracle-gap decomposition identifying timing as the dominant term.

6.7 Cross-Mechanism Check: Terra/LUNA and AMM Execution

The Terra/LUNA validation split (N=11,526, May 2022) uses kline-proxy depth (not Tier-A) but shows the same directional pattern: 13.5% price-basis > 10 bps, only 2.30% executable — a 5.9× ratio vs. 12× for SVB. The oracle earns +88 bps vs. +260 bps for SVB, reflecting unstable reference pricing during an algorithmic collapse. Both gaps persist across mechanism types.

AMM execution barrier (USDT/Curve Jun 2023, Tier-C). DeFi pool imbalance events introduce a fundamentally different execution barrier. Curve's StableSwap invariant (amplification A=2,000) compresses price deviations endogenously, keeping the AMM spot below the pool fee floor across all but the most extreme imbalances (Table 10). At peak 73%-USDT pool imbalance (Jun 16, 2023), the spot deviation is only 3.70 bps — still below the 4 bps pool fee, making AMM-route arbitrage structurally not executable even at $500K notional.

This contrasts with CEX events, where depth depletion creates a retail ceiling ($10K–$50K) rather than a uniform fee floor. Detection mechanisms also differ: Curve LP reserve imbalance accumulates gradually (enabling BOCPD early warning), whereas CEX basis is mean-reverting and yields BOCPD AUROC 0.23 (§6.5) — stress is first detectable in on-chain liquidity, not CEX prices. Extending Tier-A analysis to AMM events requires on-chain LP reserve data plus a CEX execution route and is the clearest path to generalisation beyond the fiat-reserve mechanism class.

Table 10: Curve StableSwap AMM execution gap (A=2,000, pool fee = 4 bps, TVL $800M). Spot deviation remains below the fee floor until extreme imbalance; computed by **analyze_amm_execution_gap.py**.

Pool state	Spot dev. (bps)	Net (bps)	Exec.?
Balanced (0%)	0.00	−4.00	No
Moderate (10% drift)	1.08	−2.92	No
Peak Jun 16 (23% drift)	3.70	−0.30	No
Extreme stress	7.32	+3.32	Yes

Net = spot deviation − pool fee. 73%-USDT pool imbalance ⇒ 23% drift (peak observed Jun 2023). Arbitrage requires net > 0; pool fee acts as a hard floor that depth does not.

7 LIMITATIONS AND USAGE GUIDE

Single Tier-A event. The execution-gap result rests on one stress event (USDC/SVB). Terra/LUNA provides a directional cross-mechanism check (§6.7); AMM events introduce a different execution barrier (fee floor vs. depth ceiling). Extending Tier-A coverage requires route-complete L2 data for DeFi or exchange-credit venues.

Kline-proxy buy leg. The BTC-USDC buy leg uses kline-synthesised depth for SVB (§3.1). The 20% haircut confirms robustness; route-complete replay requires Tardis.

Settlement and model scope. CEX-to-CEX latency is a fixed penalty proxy. The model ladder covers tabular and sequence ML. RL-based execution policies that act on the first detectable stress signal are the most direct extension: the oracle-gap decomposition identifies timing as the dominant component, so an agent that learns to enter at depth-peak rather than basis-peak has the clearest path to positive net returns.

Usage guide. The repository provides the full pipeline: `scripts/run_experiments.py` runs the model grid; `scripts/make_paper_figures.py` regenerates all figures; `scripts/run_shap_analysis.py` computes SHAP values; `scripts/run_meta_labeling_crossmech.py` runs the cross-mechanism meta-labeling fix; `scripts/analyze_amm_execution_gap.py` computes Curve AMM execution gaps. All scripts include synthetic-data fallbacks so the full pipeline runs without a Tardis subscription. Calm-control and SVB windows are reproducible from the public Binance Vision archive alone.

Table 11: Cross-mechanism comparison.

Metric	USDC/SVB (Tier-A)	Terra/LUNA (val.)
Minutes	15,832	11,526
Mechanism	Fiat-reserve	Alg./reflexive
Price > 10 bps	34.3%	13.5%
Executable	2.88%	2.30%
P-to-E ratio	12×	5.9×
Oracle bps	+260 bps	+88 bps†

†Kline-proxy labels; directional only.

8 CONCLUSION

Stablecoin StressBench documents a 12× price-to-execution gap in the USDC/SVB stress window: 34.3% of minutes exceed 10 bps basis on price (Layer 1); only 2.88% remain executable after costs (Layer 2). Of those 456 executable minutes, cross-mechanism meta-labeling — the paper's primary positive result — identifies opportunities capturing 50.8% of oracle net bps via 397 trades (Layer 3 under cross-mechanism conditions); all models trained on calm-control data identify 0%. At $10K notional the oracle accumulates approximately $51K in gross profit across 315 trades in the 11-day SVB window — a level of return available only to sub-$50K participants given the depth ceiling. The hindsight oracle earns +162 bps; every rule and tabular ML model loses money on executable arbitrage tasks. SHAP analysis identifies depth-imbalance as the dominant microstructure signal; false positives stem from route-direction mismatch, not thin books. GRU and MLP-Window evaluated on real Tier-A L2 data achieve −239.0 bps and −340.7 bps respectively: sequence models trained on calm-control data with zero positive stress examples do not close the gap (§6.4). AMM events introduce a qualitatively different execution barrier (fee floor rather than depth ceiling, Table 10).

Oracle gap decomposition. The 260 bps gap on `executable_arb_q10000_5m` decomposes into three derived components: (1) False-positive cost (−71.4 bps per oracle window): 581 FP trades at −38.7 bps each total −22,484 bps; spread across 315 oracle windows this is −71.4 bps per window — the single largest quantifiable drag. (2) Timing error (estimated 40–80 bps): entering 1–2 minutes late shifts execution to a lower-depth regime; the depth-withdrawal signature in the 30-minute pre-window is the primary signal a future model must capture earlier. (3) Sizing error (estimated 40–100 bps): executable rate falls from 2.88% at $10K to 1.62% at $50K, pricing out 44% of windows above the retail ceiling. Closing the gap requires models that combine cross-mechanism stress training with earlier timing, rather than larger notional. Cross-mechanism transfer (§6.5) is the empirically validated path; RL-based execution policies that act on the first detectable stress signal are the direct next step.

Data availability. The anonymised repository (URL redacted for review) contains all result CSVs, figures, YAML configs, and Python pipeline code. The Binance Vision public archive covers calm-control and SVB windows without a paid subscription. Route-complete L2 for all legs requires a Tardis subscription and is not committed to the repository.

ACKNOWLEDGEMENTS

Anonymised for review.

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